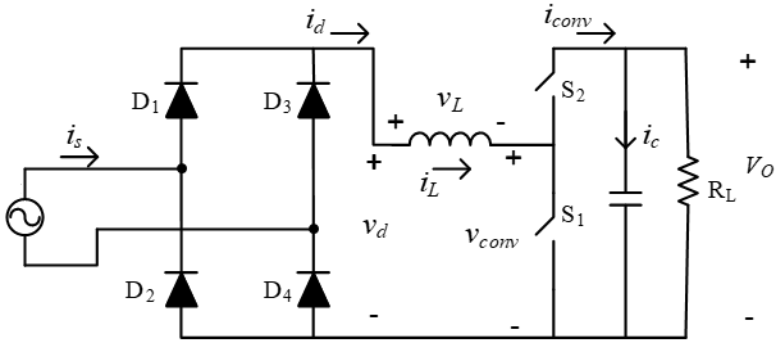
WEEK 2: ASSIGNMENT

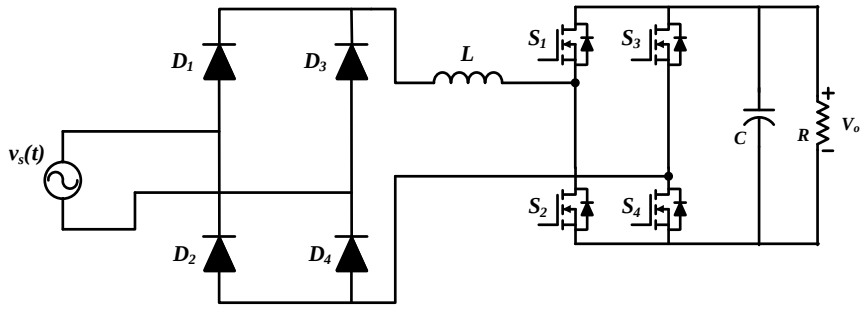
1. For the boost PFC converter shown below, what will be the voltage across the inductor during $d(t)T_s$ time interval



A. $|V_s(t)|$
B. $|V_s(t)| - V_o$
C. V_o
D. $-|V_s(t)|$
E. $-(|V_s(t)| - V_o)$
F. $-V_o$

Answer: A

2. . For the circuit shown below



The switches S_1 and S_4 are turned-on at the same time for $d(t)T_s$, while S_2 and S_3 are turned-on at the same time for $(1 - d(t))T_s$. S_1 and S_2 are switched in complementary manner; S_3 and S_4 are switched in complementary manner; the conduction time of S_1 and S_4 is always greater than the conduction time of S_2 and S_3.

What will be the expression of duty ratio ($d(t)$). Symbols have obvious meanings.

A. $1 - \frac{|v_s(t)|}{V_o}$
B. $1 + \frac{|v_s(t)|}{V_o}$

C. $\frac{\left(1-\frac{|v_s(t)|}{V_o}\right)}{2}$

D. $\frac{\left(1+\frac{|v_s(t)|}{V_o}\right)}{2}$

Answer: D

3. For the single-phase boost PFC converter, how many times the duty ratio of switches will have the same value in one line cycle

A. 1
B. 2
C. 3
D. 4

Answer: D

4. The single-phase boost PFC converter fed from 230V, 50Hz main and generates 400V at its output, then at what value of $|V_s(t)|$ the inductor current ripple will be maximum

A. 400
B. 325
C. 200
D. 230

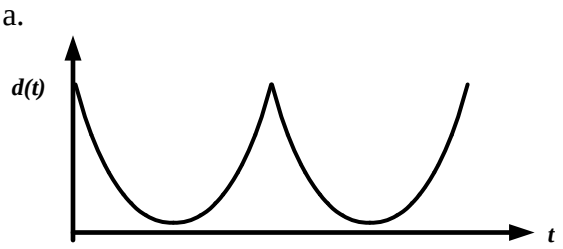
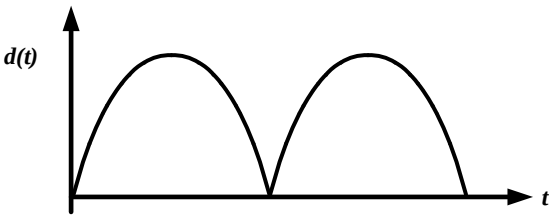
Answer: C

5. What will be the peak repetitive reverse voltage of diode in case of single-phase full-bridge rectifier with capacitive filter. Note: $V_{s,pk}$ is the peak value of input AC voltage

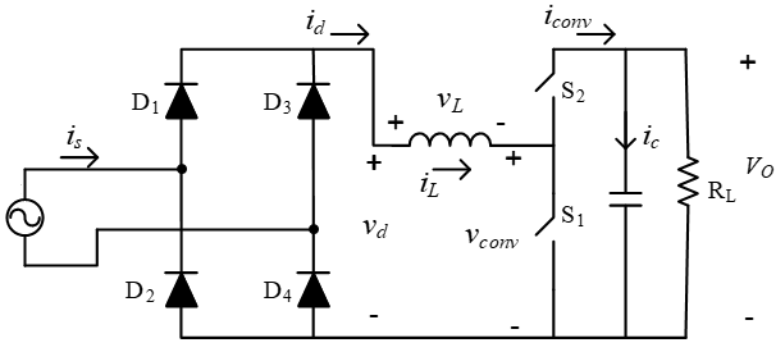
A. $V_{s,pk}$
B. $\frac{V_{s,pk}}{2}$
C. $\frac{V_{s,pk}}{\sqrt{2}}$
D. $2 * V_{s,pk}$

Answer: A

6. For the table below

Switch	Duty ratio variation
I. S_1	a. 
II. S_2	b. 

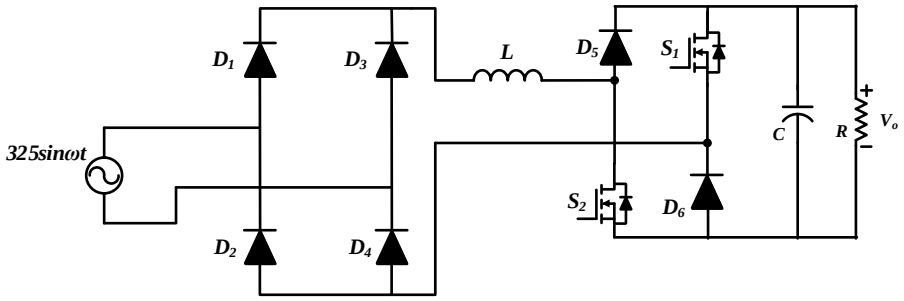
What will be the correct combination of the duty ratio variation of the switches of boost PFC converter shown below



A. I-a, II-b
B. I-b, II-a
C. I-a, II-a
D. I-b, II-b

Answer: A

7. For the circuit shown below



The switches S_1 and S_2 is turned-on for $d(t)T_s$, while D_5 and D_6 is turned-on at the same time for $(1 - d(t))T_s$. S_1 and S_2 are switched in the same time; the conduction time of S_1 and S_2 is always smaller than the conduction time of D_5 and D_6.

If the minimum value of duty ratio ($d(t)$) at positive peak of input AC voltage is 0.15, then what will be the output voltage. (Note: round the results to 2 decimal points and symbols have obvious meaning)

Answer: 463.00 to 465.00

8. A boost PFC converter fed from 230V, 50Hz input is supply the 1100W load. The converter is operated with the switching frequency of 60 kHz and regulates the output voltage at 450V. The converter is operating in continuous conduction mode, then calculate the inductance value in mH, if the allowable peak to peak current ripple is 10% of the input peak current. Assume converter is lossless converter. Note: The answer should be rounded up to 2 decimal places.

Answer: 2.70 to 2.80

9. A single phase full-bridge diode rectifier with capacitive filter is fed from 220V, 50Hz mains supply. A 200W load is connected to the output cannot take a voltage ripple of more than 10% of the peak value of the input. Calculate the smallest value of capacitance (in µF) that should be used in order to meet the requirement. (Note: The answer should be rounded up to 2 decimal places; use $\pi = 3.14$; $1\ radian = 57.3\ degrees$).

Answer: 185.00 to 190.00

10. For the single-phase boost PFC converter fed from 50 Hz AC source having varying voltage 180 V-265 V. What will be the minimum value of duty ratio ($d(t)$) of switch S_1 if the output voltage is kept at 500V. (Note: round the result to 2 decimal points)

Answer: 0.23 to 0.27

Answer Key:

1. A	2. D
3. D	4. C
5. A	6. A
7. 463.00 to 465.00	8. 2.70 to 2.80
9. 185.00 to 190.00	10. 0.23 to 0.27